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pyTMD: Python-based tidal prediction software

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Summary

pyTMD is an open-source tidal prediction software that aims to simplify the calculation of ocean and Earth tides. It is designed to be able to handle a wide range of model formats and can incorporate different physics schemes. This flexibility allows pyTMD to be tailored to fit specific use cases, while still allowing ease of use.

Statement of need

pyTMD is a generalized tide program that allows users to calculate both tide deflections and currents. As a tide model driver, pyTMD can read from a broad suite of models, and use different physics schemes in the internal calculations. Over 50 different models are presently supported within pyTMD, and additional models can be defined with a JSON file. The online documentation contains background information for tide modeling and multiple Jupyter notebooks that demonstrate core functionality.

pyTMD was designed to be used by beginners and scientific researchers alike. It has been used in a number of scientific publications for modeling regional tides (Freer et al., 2023; Millan et al., 2023; T. C. Sutterley et al., 2019), modeling global tides (Gregg et al., 2024; Paprotny et al., 2024), and creating several Earth observation datasets (ENVEO et al., 2021; Smith et al., 2024a, 2024b). It has also been leveraged within larger earth-observation software packages (Bishop-Taylor et al., 2025; Fitzpatrick et al., 2024).

Functionality

Ocean and Load Tides

Tides are frequently decomposed into harmonic constants, or constituents, associated with the relative positions of the sun, moon and Earth (Padman et al., 2018). For ocean and load tides, `pyTMD.io` contains routines for reading major constituent values from commonly available tide models, which typically fall within a few general formats: `OTIS-binary` (Egbert & Erofeeva, 2002; Padman et al., 2008), `OTIS-compact`, `OTIS-netcdf`, `TMD3-netcdf` (Greene et al., 2024), `GOT-ascii` (Ray, 1999), `GOT-netcdf`, `FES-ascii` (Le Provost et al., 1994) and `FES-netcdf`. Information for each of the supported tide models is stored within a JSON database. `pyTMD.io` reads or computes the amplitude and phase lag of tide model constituents, which are then interpolated to the output spatial coordinates.

pyTMD uses the astronomical argument formalism outlined in Doodson & Lamb (1921) to predict ocean and load tides. After the computation of the amplitude and phase lag for the spatial coordinates, pyTMD computes the temporal elements using by 1) calculating the

astronomical angles (S, H, P, N, P_s) (Meeus, 1991; Simon et al., 1994), 2) combining these angles with the "Doodson numbers" in a Fourier series to compute the equilibrium tide phase (G) of each constituent, and 3) computing the 18.6-year nodal amplitude and phase corrections (f and u) of each constituent (Dietrich, 1980; Doodson & Lamb, 1921; Pugh & Woodworth, 2014). The spatial and temporal components are then combined, and the output tidal time series is calculated through a summation over all constituents (Egbert & Erofeeva, 2002). The contributions of "minor" constituents can then be estimated using inference methods (Ray, 2017; Schureman, 1958).

Separately, long-period ocean tides can be predicted assuming an "equilibrium response" (Cartwright & Edden, 1973; Cartwright & Tayler, 1971; Doodson & Lamb, 1921). Here, the oceanic surface is estimated to respond instantaneously to the tide-producing forces of the moon and sun, and is not influenced by inertia, currents or the irregular distribution of land (Proudman, 1960; Ray & Erofeeva, 2014; Schureman, 1958).

pyTMD has baseline capability to decompose a time series into tidal constants using the harmonic method outlined in M. G. Foreman & Henry (1989) and M. G. G. Foreman et al. (2009). The 18.6-year nodal modulations are taken into account when building the design matrix for the least-squares solver (Egbert & Erofeeva, 2002; M. G. G. Foreman et al., 2009):

$$h(t) = z_0 + \sum_{k=1}^{n} f_k(t) \left[X_k \cos\left(G_k(t) + u_k(t)\right) - Y_k \sin\left(G_k(t) + u_k(t)\right) \right] + \sigma$$

where $h(t)$ is the sea level height at time t, z_0 is mean sea level, $f_k(t)$ is the (nodal) amplitude modulation in constituent k, $G_k(t)$ is the phase of the equilibrium tide, $u_k(t)$ is the (nodal) phase correction angle, and σ is the uncertainty. The amplitude A_k and phase θ_k of the constituent are calculated using the solution coefficients X_k and Y_k (Egbert & Erofeeva, 2002):

$$A_k = \sqrt{X_k^2 + Y_k^2}$$
$$\theta_k = \tan^{-1}\left(\frac{-Y_k}{X_k}\right)$$

Pole Tides

Load and ocean pole tides are driven by variations in the Earth's rotation axis with respect to its mean position, along with corresponding elastic responses and secondary effects (Desai, 2002; Desai et al., 2015; Wahr, 1985). `pyTMD.predict` estimates pole tide variations using polar motion values from the IERS "finals" file, and the reference "secular" pole position calculated following the latest IERS conventions (Petit & Luzum, 2010).

Solid Earth Tides

The tidal deformation of the solid Earth is modeled in pyTMD using the Love and Shida number formalism described in Wahr (1981) and Mathews et al. (1997) and formalized in the IERS Conventions (Petit & Luzum, 2010). `pyTMD.astro` has options for calculating approximate ephemerides following Meeus (1991) and Montenbruck (1989) or using high-resolution JPL ephemerides from Park et al. (2021) using the `jplephem` package (Rhodes, 2011). The calculation for solid Earth tides includes the frequency-dependent and dissipative mantle anelasticity corrections from Mathews et al. (1997).

Time

pyTMD uses the `timescale` library (T. Sutterley et al., 2025) to manage temporal conversions, calculate "dynamical" time scales, and calculate Earth Orientation Parameters (EOPs).

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